

The effectiveness of parental distraction during children's acute pain: The moderating effect of socioeconomic status

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Abstract

Background: Parental responses to children's pain shape how children interpret and cope with pain symptoms through parental modelling and operant conditioning. Evidence suggests that parental distraction is effective in reducing children's acute pain responses, but findings are inconsistent across pain tolerance, intensity and unpleasantness, and are limited to samples of primarily middle and upper-middle class families. Although socioeconomically disadvantaged families may have fewer psychological resources to cope with pain, no studies have examined whether the utility of parent distraction varies by family socioeconomic status (SES). The current study tested the hypothesis that relations between parental distraction and acute pain responses in children vary by family SES, with children from higher versus lower SES families experiencing more substantial benefits.

Methods: Children's pain symptoms and parents' use of verbal distraction during a cold pressor task were examined in a community sample of 530 twin children aged 7–12 years old and their primary caregivers.

Results: Parental distraction was positively associated with children's pain tolerance and unrelated to intensity and unpleasantness, but these associations are qualified by significant moderation. In families with higher SES, parental distraction was an effective technique in children's pain management, associated with more pain tolerance and less pain intensity and unpleasantness. However, for families with lower SES, these same benefits were not present.

Conclusions: Findings suggest that the effectiveness of parental distraction for children's acute pain may depend on family SES.

Significance: Study findings suggest that the effects of parental distraction on children's responses to an acute pain task vary by family SES. Although parental distraction may be effective for higher SES children, further research is needed to identify whether and why distraction may not be beneficial for lower SES families.

1 | INTRODUCTION

Environmental context influences children's pain experiences (Palermo & Chambers, 2005; Palermo, Valrie, & Karlson, 2014). In particular, parental responses to children's pain have been linked with children's acute pain

symptoms (Chambers, Craig, & Bennett, 2002; Moon, Chambers, & McGrath, 2011) and coping behaviours (Frank, Blount, Smith, Manimala, & Martin, 1995). Theory suggests that children learn how to interpret and manage pain symptoms through parental modelling and operant conditioning (Cordier & Diers, 2018; Fordyce, 1977; Prkachin

& Craig, 1986). Accordingly, pain-reducing parenting is thought to involve directing the child's attention away from the pain (e.g. distracting), whereas pain-promoting parenting involves pain-directed talk and attention (Manimala, Blount, & Cohen, 2000); however, these associations may depend on sociocultural factors (Evans et al., 2008).

Evidence points to the utility of parent-provided distraction versus attention during acute pain episodes for lowering distress among healthy children (Gonzalez, Routh, & Armstrong, 1993; Manimala et al., 2000; Walker et al., 2006). Parental distraction has also been linked with better pain tolerance and fewer child-rated pain symptoms, but it is unclear whether distraction works similarly for children across SES backgrounds. Observed maternal and paternal verbal distraction versus verbal attention/empathy during a cold pressor task (CPT; pain induction by cold water) was associated with higher pain tolerance, and maternal distraction was associated with lower pain unpleasantness, but not intensity, for healthy children aged 8–12 years primarily of European descent from middle and upper-middle class backgrounds (Moon et al., 2011). When mothers were randomized to behave as usual (control) or in ways to promote (e.g. verbal attention/empathy) versus reduce (e.g. verbal distraction) pain, girls but not boys in the pain-reducing condition reported lower pain intensity, but not unpleasantness or tolerance (Chambers et al., 2002). Thus, children's pain intensity ratings may be more susceptible than their psychophysiological responses to temporary changes in parental behaviour during a pain-induction task. Additional research using ecologically valid methods (i.e. in-home observation) with socioeconomically diverse samples is needed to evaluate the utility of parental distraction for children's acute pain responses and whether it is generalizable.

Lower versus higher SES parents are more likely to experience chronic stress and reduced capacity to provide supportive and sensitive parenting (Conger, 1994; McLoyd, 1990), which may impact the quality of the pain-related distraction they provide to their children. Additionally, children from lower versus higher SES families may have greater emotional dysregulation (Kim et al., 2013) and fewer psychological resources, like optimism and positive coping strategies, to help them manage perceived threats such as pain (Finkelstein, Kubzansky, Capitman, & Goodman, 2007). Although children from lower versus higher SES families more frequently report pain symptoms, including headache (Bugdayci, 2005; Grøholt, 2003) and stomach ache (Kristjansdottir, 1996), it is unclear whether these children have greater difficulty using coping strategies to manage acute pain symptoms.

In the present study, twin children separately participated in a CPT conducted in the home with their primary caregiver present to evaluate the hypotheses that: 1) parental verbal distraction predicts higher child acute pain tolerance, lower intensity and lower unpleasantness, and 2) benefits of parental

distraction for child pain responses are stronger for higher versus lower SES families.

2 | MATERIALS AND METHODS

2.1 | Participants and procedure

Participants were primary caregivers ($N = 265$; 94.33% mothers, 5.67% fathers) and their school-aged twin children ($N = 530$; age range: 7–12 years) who were enrolled in a larger ongoing longitudinal study, the Arizona Twin Project (Lemery-Chalfant et al., 2019). Participants were either recruited from state birth records or through online advertising and represent families from both rural and urban parts of the southwestern United States. Twins are representative of typically developing children for a wide range of physical and mental health outcomes (Kendler & Prescott, 2007). Although twin studies are useful for estimating genetic and environmental influences on trait variation, this method was not utilized to answer the current research questions.

All study procedures were reviewed and approved by the Institutional Review Board at Arizona State University prior to the start of data collection. All primary caregivers provided consent prior to begin data collection and child assent was obtained during study home visits before administration of study procedures. Prior to study home visits, primary caregivers reported on demographic and socioeconomic information through an online survey or by telephone interview. During study home visits, children participated in a filmed CPT with their primary caregiver present.

The characteristics of the cold pressor device, water temperature and procedures used in this study are in line with published guidelines (von Baeyer, Piira, Chambers, Trapanotto, & Zeltzer, 2005). The device used is a plastic lidded cooler with a 4.5" diameter opening. A plastic panel separates the cooler into two sections, allowing for a chamber to hold ice and a pump to circulate water throughout the tank, and a chamber for hand immersion. Prior to the task, water was cooled with ice to $50^{\circ}F \pm 1.8^{\circ}F$ ($10^{\circ}C \pm 1^{\circ}C$). Children were excluded from participation if they had Raynaud's disease or current injuries on both hands, as reported by their parents prior to the task.

During study home visits, each twin child participated in separate study activities in separate rooms. Therefore, the children did not have the opportunity to observe their twin's performance on the CPT. During the CPT, children were seated next to the cold pressor device, with children randomized to use their dominant or non-dominant hand for the task. Children sat across from their parent, with approximately 18 inches of separation between chairs, with video cameras positioned to record each individual's face and body. Next, a trained research assistant placed a gauze band around the child's arm to indicate how

deep the arm was to be placed in the device to ensure hand immersion up to the wrist. After describing the device and showing the interior of the cooler to the participants, the research assistant instructed the child to keep their hand in the water for as long as possible, but to remove it if it became too uncomfortable or too painful. Parents were instructed to interact with their children as they normally would. If the child kept their hand in the water until the maximum time of 240 s, the research assistant instructed the child to remove it, in line with published guidelines (von Baeyer et al., 2005). After completion of the task, children were instructed to refrain from discussing their experience with the CPT with their sibling until the end of the home visit.

2.2 | Measures

2.2.1 | Socioeconomic status (SES)

SES was assessed with an open-ended question on a survey that asked parents to report their estimated total family income for the past year, prior to taxes. Participants were asked to provide a value based on all sources of income, including jobs, self-employment, government assistance, disability, child support and social security for all adults who contributed to household expenses.

2.2.2 | Parental use of distraction

Distracting behaviour was measured by the proportion of time during the child's engagement in the CPT that parents spent verbally directing their child's attention away from the experience and task at hand. Trained research assistants coded the video recordings for presence or absence of parental verbal distraction behaviour every 10 s for the duration of the task, adapting the coding scheme developed by Moon et al. (2011). All coders reached reliability after extensive training and completion of a reliability set at or above 90% agreement with the master coder. The master coder independently coded 10% of all of the videotapes and research assistants continued to maintain reliability at or above 90% with the master coder.

The task started once the child placed their hand in the water and ended either when the child removed their hand or when the child reached the maximum time of 240 s. Verbal distraction behaviour was coded if the content of parent-initiated speech was unrelated to the task or the experience of the task. Examples of distracting verbal behaviour include the parent telling a joke or story; engaging the child in song, play or counting; discussing future plans or past events; discussing an object in the room; and asking the child what they would like to talk about to take their mind off of the pain. Proportion of verbal distracting behaviour (i.e. parental distraction) was

calculated by dividing the number of 10-s segments in which there was verbal distraction by the total number of segments.

2.2.3 | Acute pain tolerance

Acute pain tolerance was measured by the number of seconds children kept their hand in the water during the CPT. Research assistants used a stopwatch to start the time as soon as the child's hand lined up with the depth specified by a gauze band around their arm. Time was stopped as soon as the child began to remove his/her hand, or when the duration reached 240 s. Using the twin design, the heritability of pain tolerance during the CPT with this sample was 65% (Lemery-Chalfant, Davis, Clifford, & Kaur, 2020), suggesting that this measure taps a heritable trait, with children's differences also influenced by the environment.

2.2.4 | Acute Pain intensity

Child acute pain intensity was assessed immediately after the child completed the CPT. Trained research assistants asked participants to rate the worst pain felt during the CPT by selecting one of the six pain faces representing values 0 (no pain) to 10 (very much pain) from the Faces Pain Scale-Revised (FPS-R; Hicks, von Baeyer, Spafford, van Korlaar, & Goodenough, 2001). The scale has demonstrated concurrent validity with visual and coloured analogue scales in children aged 4–12 years (Hicks et al., 2001).

2.2.5 | Acute pain unpleasantness

Child acute pain unpleasantness was assessed immediately after the child completed the CPT using the Facial Affective Scale (FAS; McGrath, 1985). Trained research assistants asked participants to choose one of the nine faces representing how they felt during the CPT. Values were assigned to each face, with the lowest value (0.04) representing the happiest face possible and the highest value (0.97) representing the saddest face possible. In one study, the scale had good concurrent validity with a coloured analogue scale of pain unpleasantness in children aged 4–10 years, and appeared to measure a distinct construct in relation to a measure of pain intensity, the Faces Pain Scale (Goodenough, Van Dongen, Brouwer, Abu-Saad, & Champion, 1999).

2.3 | Analytic strategy

Multilevel modelling was used for main study models to account for the hierarchical nature of the data, with children

TABLE 1 Means, Standard Deviations and Intercorrelations for Main Study Variables

Variable	<i>M</i>	<i>SD</i>	Range	Intercorrelations			
				2	3	4	5
1 Distraction	0.40	0.41	0–1	0.12*	0.21**	–0.08	–0.08
2 Income (\$)	110,800	85,639	5,000–600,000	–	0.17**	–0.12**	0.07
3 Acute pain tolerance (sec)	92.69	90.94	5–240	–	–	–0.31**	–0.32**
4 Acute pain intensity	4.84	2.91	0–10	–	–	–	0.61**
5 Acute pain unpleasantness	0.61	0.26	0.04–0.97	–	–	–	–

* $p < .05$; ** $p < .01$.

clustered within families. Models examined relations between parental use of distraction and children's acute pain tolerance, intensity and unpleasantness. Additional models examined whether SES moderated main effects, such that parental use of distraction, SES, and the interaction between distraction and SES were tested as predictors separately for pain tolerance, intensity and unpleasantness. Significant interactions were probed using simple slopes for SES at one standard deviation above, at and below the mean for SES. Standardized *z*-score values of income were used for SES to adjust for scaling. The outcome variable of child acute pain tolerance was log transformed to ensure normality of the multilevel regression residuals, in line with published guidelines (Cohen, Cohen, West, & Aiken, 2003). Model predictors, including parental use of distraction, SES and their interaction, were grand-mean centred. Child sex (0 = boys, 1 = girls), ethnicity (0 = Caucasian, 1 = other) and child age were tested as covariates and retained in models if significant. Descriptive statistics and intercorrelations among main study variables did not account for dependence of children clustered within families. Analyses were conducted using SPSS Version 24.

3 | RESULTS

3.1 | Sample characteristics

Twin children (29.39% monozygotic, 70.61% dizygotic) were evenly represented by sex (51.70% male), were on average 9.70 years old ($SD = 0.80$, range = 7.77–12.01), and were from diverse racial and ethnic backgrounds (58.91% Non-Hispanic White, 24.29% Hispanic/Latinx, 4.86% African American, 2.43% Native American, 5.21% multi-racial, 4.30% other). Primary caregivers were predominantly mothers (94.33%), married (81.46%, 6.46% divorced, 5.00% always single, 3.33% partnered, 1.67% widowed, 1.25% separated, 0.83% other), were on average 39.65 years old ($SD = 5.85$), and were racially and ethnically diverse (68.62% Non-Hispanic White, 21.48% Hispanic/Latinx, 2.83% African American, 2.83% Native American, 2.42% multi-racial, 1.82% other). Total annual household income was on average \$110,800.00

($SD = \$85,639$), with approximately two-thirds of the sample reporting between \$25,162 and \$196,439. The median for primary caregiver educational attainment was a 4-year bachelor's degree, with two-thirds of the sample reporting at least some college and up to 1–2 years of graduate school.

3.2 | Descriptive statistics and intercorrelations

Table 1 displays means, standard deviations and intercorrelations for primary study variables. All children kept their hand in the water for a minimum of 5 s and 25.70% of children kept their hand in the water for the maximum time of 240 s. The average amount of time parents spent distracting was 40.19% ($SD = 41.02\%$). Over half the parents (58.95%) engaged in some amount of distracting throughout the task, with 27.10% of parents providing distraction within the first 10 s of the task. Higher SES was modestly associated with more frequent distracting behaviours. SES was positively associated with children's acute pain tolerance, whereas it was negatively related to children's acute pain intensity, and unrelated to children's pain unpleasantness. Children's pain tolerance was negatively related to their pain intensity and unpleasantness, and the latter two measures were moderately and positively correlated.

3.3 | Prediction of acute pain tolerance, pain intensity and pain unpleasantness

Table 2 displays the findings of multilevel regression models examining the prediction of acute pain tolerance, pain intensity and pain unpleasantness by parental distraction. Covariates (i.e. child sex, child ethnicity and child age) were tested and retained in models if they were significant. Parental distraction was related to child pain tolerance, such that higher parental distraction was related to higher child pain tolerance during the CPT. Child sex and age were also related to child pain tolerance, such that girls and older children had higher pain tolerance than boys and younger children during

the task. Parental distraction was not associated with child acute pain intensity or acute pain unpleasantness. Child sex was related to acute pain intensity, such that girls had lower acute pain intensity during the CPT than boys. Covariates (i.e. child sex, age and ethnicity) were tested as moderators of relations between parental distraction and pain outcomes, but they were not significant.

Table 3 presents the findings of multilevel regression models examining estimates for all three pain outcomes as a function of parental distraction, SES and the interaction of parental distraction and SES. The interaction of parental use of distraction and SES was significant for acute pain tolerance, acute pain intensity and acute pain unpleasantness, such that relations between parental distraction and outcomes varied as a function of SES. Interaction effects were probed by examining simple slopes for parental distraction on outcomes separately for lower (1 *SD* below the mean, \$25,162), middle (mean, \$110,800) and higher (1 *SD* above the mean, \$196,439) SES families in the given sample, in accordance with guidelines described by Aiken and West (1991). Results are presented in Table 4.

There was a significant positive association between parental distraction and acute pain tolerance for the higher and middle SES families, but not for the lower SES families (see Figure 1a). There was a significant negative association between parental distraction and acute pain intensity for the higher SES families, but not for the middle and lower SES families (see Figure 1b). The relation between parental distraction and pain unpleasantness was negative for the higher SES families, positive for the lower SES families and

TABLE 2 Results of multilevel regression analyses predicting acute pain tolerance, intensity and unpleasantness from parental distraction ($N = 530$)

Variable	Estimate	Std. Error	<i>p</i> -value
Outcome: Acute pain tolerance			
Time distracting [†]	0.228	0.050	<.01
Child sex	0.133	0.037	<.01
Child age	0.063	0.029	<.01
Outcome: Acute pain intensity			
Time distracting	−0.615	0.341	.08
Child sex	−0.621	0.249	.02
Outcome: Acute pain unpleasantness			
Time distracting	−0.049	0.031	.11

Note: [†]Time Distracting was measured by the proportion of time parent spent distracting during the task. Time Distracting was grand-mean centred. Log-transformed values for acute pain tolerance were used. Child sex (0 = boys, 1 = girls), ethnicity (0 = Caucasian, 1 = other) and child age were tested as covariates and retained in models if significant.

TABLE 3 Results of multilevel regression analyses predicting acute pain tolerance, intensity and unpleasantness from parental distraction, ses and parental distraction X SES ($N = 530$)

Variable	Estimate	Std. Error	<i>p</i> -value
Outcome: Acute pain tolerance			
Time distracting [†]	0.220	0.053	<.01
SES	0.020	0.025	.42
Time distracting X SES	0.210	0.054	<.01
Child sex	0.118	0.039	<.01
Outcome: Acute pain intensity			
Time distracting	−0.462	0.365	.21
SES	−0.131	0.167	.41
Time distracting X SES	−1.135	0.370	<.01
Child sex	−0.720	0.282	.02
Outcome: Acute pain unpleasantness			
Time distracting	−0.025	0.032	.44
SES	0.004	0.015	.80
Time distracting X SES	−0.130	0.033	<.01

Note: [†]Time Distracting was measured by the proportion of time parent spent distracting during the task. Time Distracting, SES and their interaction were grand-mean centred. Standardized values of SES and log-transformed values of acute pain tolerance were used. Child sex (0 = boys, 1 = girls), ethnicity (0 = Caucasian, 1 = other) and child age were tested as covariates and retained in models if significant.

unrelated for the middle SES families (see Figure 1c). Child sex was accounted for in models predicting acute pain tolerance and intensity, in line with previous analyses.

3.4 | Exploratory analyses: Variability in parental distraction across twins

Post hoc analyses were conducted to examine parents' variability in providing distraction across their twin children. Because lower versus higher SES parents face increased barriers in providing sensitive parenting (Conger, 1994; McLoyd, 1990), it is possible that findings may be explained by lower SES parents' reduced variability in providing individually tailored distraction to their children. Difference scores were created to represent the amount of variability in parental distraction given to each of the two children, ranging from 0 (no difference in proportion of time spent distracting between twins) to 1 (total difference). For parents who completed the task with both twin children ($N = 236$), 44.10% spent the same proportion of time during the CPT providing distraction to both children, with an average absolute difference score of 0.18 ($SD = 0.25$). The distraction difference score was not correlated with family SES ($r = -0.04$,

TABLE 4 Results of multilevel regression analyses of parental distraction on acute pain tolerance, intensity and unpleasantness at lower, middle and higher income ($N = 530$)

Variable and analysis	Simple slope	Estimate	Std. error	<i>p</i> -value
Outcome: Acute pain tolerance				
Lower income	$\hat{y} = 0.010x + 1.647$	0.010	0.074	.90
Middle income	$\hat{y} = 0.220x + 1.667$	0.220	0.053	<.01
Higher income	$\hat{y} = 0.430x + 1.688$	0.430	0.076	<.01
Outcome: Acute pain intensity				
Lower income	$\hat{y} = 0.673x + 5.491$	0.673	0.515	.20
Middle income	$\hat{y} = -0.462x + 5.353$	-0.462	0.365	.21
Higher income	$\hat{y} = -1.597x + 5.215$	-1.596	0.524	<.01
Outcome: Acute pain unpleasantness				
Lower income	$\hat{y} = 0.105x + 0.615$	0.105	0.045	.03
Middle income	$\hat{y} = -0.025x + 0.619$	-0.025	0.032	.44
Higher income	$\hat{y} = -0.155x + 0.623$	-0.155	0.046	<.01

Note: Standardized values of SES and log-transformed values of acute pain tolerance were used.

Lower income = 1 standard deviation (*SD*) below mean for income = \$25,162; Middle income = at the mean for income = \$110,800; Higher income = 1 *SD* above the mean for income = \$196,439. Child sex (0 = boys, 1 = girls) was included in models predicting acute pain intensity and tolerance in line with main analyses.

$p = .60$), which indicates that parents' variability in providing distraction to their twin children during the CPT was not associated with family SES.

4 | DISCUSSION

This study examined relations between parental use of verbal distraction during a CPT and children's acute pain tolerance, intensity and unpleasantness in a socioeconomically and ethnically diverse sample of twin children. Higher parental distraction was associated with higher acute pain tolerance, but was unrelated to ratings of acute pain intensity and unpleasantness. Moreover, relations between parental use of distraction and children's acute pain tolerance, intensity and unpleasantness varied as a function of SES, such that the benefits of parental distraction accrued primarily to children from higher SES families. The current findings highlight that the value of parental distraction for acute child pain varies as a function of pain outcome and family SES.

We hypothesized that parental distraction would improve pain tolerance, intensity and unpleasantness, but found that parental distraction only promoted higher acute pain tolerance. These results did not vary by child demographic characteristics, contrary to findings from Chambers et al. (2002) study that experimentally manipulated maternal distraction and found that it was associated with decreased pain intensity only for girls. Findings are partially consistent with a previous study of primarily middle and upper-middle class

families that found that observed maternal and paternal distraction was associated with higher pain tolerance, and maternal distraction was associated with lower pain unpleasantness (Moon et al., 2011). Although we did not find an association with pain unpleasantness, this discrepancy in findings across studies could be attributed to differences in sample characteristics, including socioeconomic diversity. This study also sought to examine whether parental use of distraction interacted with family SES, as measured by family household income, on children's acute pain outcomes.

Consistent with our hypotheses, findings indicated that relations between parental use of distraction and children's acute pain tolerance, intensity and unpleasantness varied as a function of SES. Increased parental distraction was linked with higher pain tolerance in children from the middle and higher but not the lower SES households. Increased parental distraction was linked with lower pain intensity ratings in children from the higher but not the middle or lower SES households. Finally, for acute pain unpleasantness, increased parental distraction was associated with lower scores among children from the higher SES households, higher scores among children from the lower SES households and was unrelated to scores among children from the middle SES households. Taken together, the current results suggest that SES represents an important social contextual factor that shapes the utility of pain-related parenting strategies.

Pain-related outcomes also varied as a function of child characteristics. Older children had higher pain tolerance during the CPT compared with younger children, consistent with results from prior studies (Birnie, Parker, & Chambers, 2016;

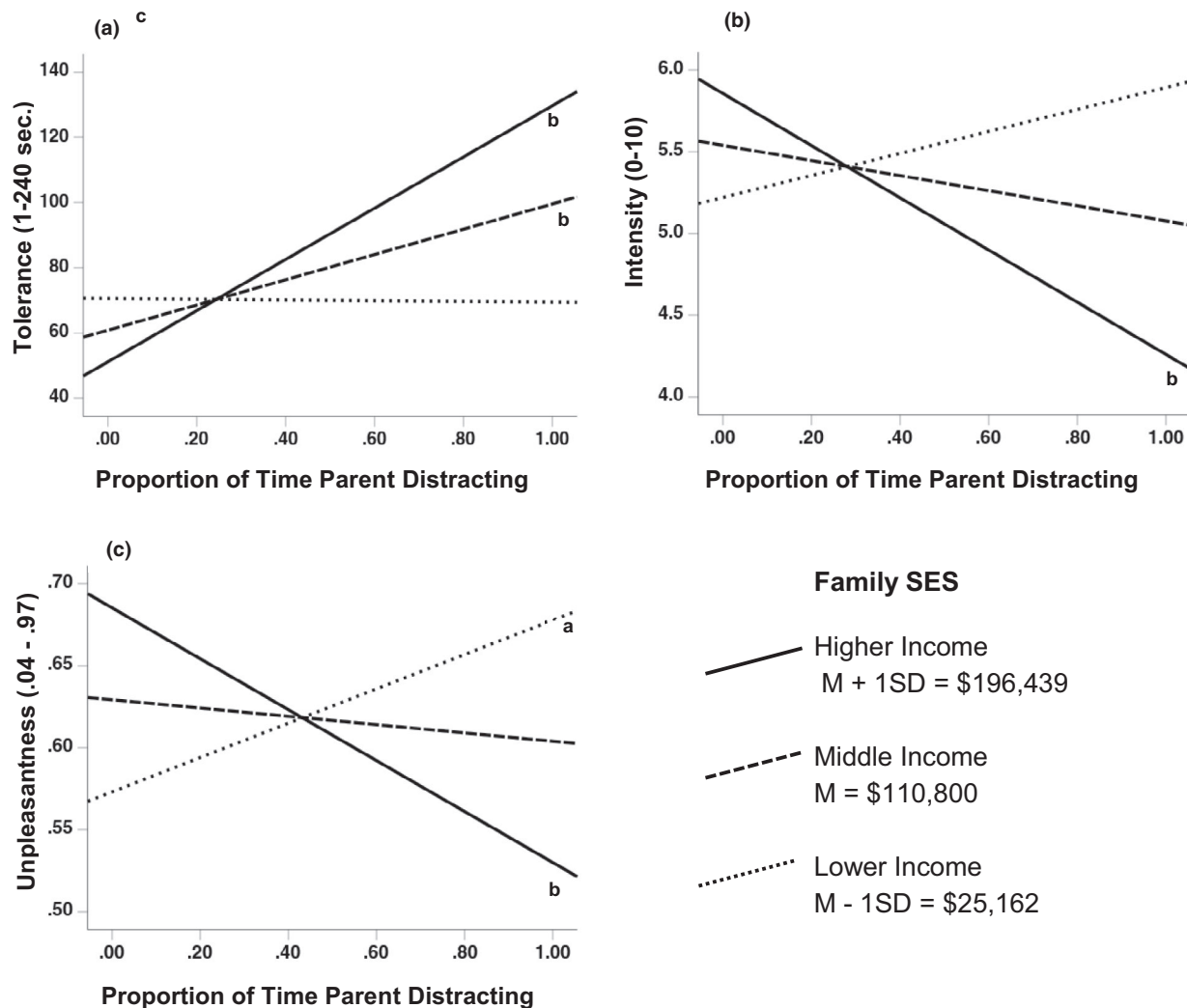


FIGURE 1 (a-c) Parental distraction during an acute pain task by children's (a) acute pain tolerance, (b) acute pain intensity and (c) acute pain unpleasantness for socioeconomic status (SES; family income level) at one standard deviation (SD) above (\$196,439), at (\$110,800) and below (\$25,162) the sample mean (M) ($N = 530$). ^a $p < .05$; ^b $p < .01$. ^cAlthough Tables 2–4 use log-transformed variable for acute pain tolerance, (a) uses untransformed variable for interpretability

Zeltzer, Fanurik, & LeBaron, 1989). Girls had higher pain tolerance and lower pain intensity than boys, contrary to findings in a meta-analysis indicating no sex differences among children under 12 years of age (Boerner, Birnie, Caes, Schinkel, & Chambers, 2014). However, it should be noted that the current study's sample is comparable in size to the entire meta-sample (Boerner et al., 2014). Furthermore, the studies included in Boerner et al., (2014) had a wider age range, with children as young as 3 years, and their sociodemographic characteristics were unknown. Therefore, differences in sample characteristics may account for disparate findings between studies.

To the best of our knowledge, this is the first study to suggest that parental use of distraction for the management of children's acute pain symptoms may not be effective for all children across the socioeconomic gradient. Although study results provide support for the use of distraction during acute pain for higher SES families, there is a lack of

evidence of the benefits of distraction techniques for lower SES families. The mechanisms underlying the moderating effects of SES on the parental distraction–pain relations among children have yet to be elaborated. One possibility is that lower SES environments may affect children's ability to use distraction during pain episodes due to the perception of pain as a threat. The exaggeration of threat value of pain stimuli, known as pain catastrophizing (Chaves & Brown, 1987), has been found to interact with distraction techniques to predict the experience of pain in both children and adults, such that distraction is less effective among those high versus low in catastrophizing (Prins, Decuyper, & Van Damme, 2014; Verhoeven, Goubert, Jaaniste, Van Ryckeghem, & Crombez, 2012). As lower SES has been linked to greater catastrophizing among adults with chronic pain (Feldman, Dong, Katz, Donnell-Fink, & Losina, 2015), it is possible that distraction techniques may be ineffective

for children of lower SES, who may be more vulnerable to stress and may have fewer psychological resources to help them cope with stressors such as acute pain episodes.

Alternatively, lower SES conditions (e.g. higher stress, lower psychological resources) may impact parents' ability to provide high-quality distraction (e.g. sensitivity to children's cues, efforts to engage the child, confidence in directing the child; Kleiber, McCarthy, Hanrahan, Myers, & Weathers, 2007). As both quantity and quality of parental distraction have been associated with pain-related distress in children (McCarthy et al., 2010), it is possible that the current study's findings may be explained by lower SES parents' reduced capacity to provide sensitive and high-quality distraction to one or both twin children. Post hoc analyses did not support the idea of differential use of distraction being dependent on SES, but this potential moderating role of parental sensitivity requires continued consideration. Future research should examine whether various aspects of distraction quality account for the possible differential benefits on child pain outcomes by SES.

This study has a number of strengths and some limitations to consider. With regard to strengths, the community sample used in the study represents participants from diverse ethnic and socioeconomic backgrounds. Additionally, the CPT was completed in participants' homes to maximize ecological validity. Behavioural observation by independent observers has been used routinely to measure parenting and the parent-child relationship (McMahon & Forehand, 2003), and has generally been found to have good social validity (Rhule, McMahon, & Vando, 2009; Sanders & Dadds, 1993). One limitation is that primary caregivers who completed the task with their children were primarily mothers, so the extent to which findings generalize to other types of caregivers' use of distraction is unknown. Further research is needed to determine whether the use of distraction is similarly related to children's acute pain symptoms and tolerance when provided by secondary caregivers, grandparents, teachers and healthcare professionals, and whether relations continue to vary by family SES. A second limitation is that few studies have examined whether differences in acute pain responses during a CPT have meaningful implications for chronic pain or other future pain-related outcomes in children. However, the CPT is regarded as an objective measure of pain sensitivity (Oaks, Stage, Middleton, Faraone, & Johnson, 2018), one indicator that may be linked to the development of chronic pain (Nielsen, Staud, & Price, 2009). Studies among adults indicate that lower pain tolerance during a CPT predicts worse pain symptoms post-surgery (Kasch, Qerama, Bach, & Jensen, 2005) and during recovery from injury (MacLachlan, Shipton, & Wells, 2016). Examining the predictive validity of pain-induction tasks on child chronic pain symptoms will help to determine whether

such measures should be used in the future to predict those most at risk for chronic pain.

In summary, findings indicate that parental use of distraction during a CPT task was related to better pain tolerance in children, but not children's ratings of acute pain intensity and unpleasantness, in a diverse community sample. Importantly, these findings were conditioned on family SES; increased parental use of distraction was linked to higher child pain tolerance and lower child pain intensity and unpleasantness among children from higher but not lower SES families. As parental responses to children's pain influence children's pain behaviours and likelihood of pain-related disability (Simons, Claar, & Logan, 2008; Walker, Garber, & Greene, 1993), clarifying the effectiveness of a commonly used technique for the management of acute child pain is vitally important. Future research should seek to both replicate findings and examine mechanisms underlying factors related to children's differential response to parent distraction. Should additional evidence suggest that children from disadvantaged backgrounds receive little benefit from parental use of distraction, further research should examine whether alternative approaches work better for the management of acute pain.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

AUTHORS' CONTRIBUTION

Shannon V. Moore, Mary C. Davis, and Kathryn Lemery-Chalfant were involved with the conception and design of the study. Shannon V. Moore and Emma K. Lecarie completed data analysis, and all authors were involved with interpretation of the data. Shannon V. Moore and Emma K. Lecarie drafted the manuscript, and Mary C. Davis and Kathryn Lemery-Chalfant revised the manuscript for intellectual thoroughness and accuracy. All authors approved of the final version to be published.

ETHICAL APPROVAL

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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